

South Asia's Water Crisis

The Indus, Ganges-Brahmaputra, and the Himalayan Glacier Emergency

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Executive Summary

South Asia's water architecture — the Himalayan glaciers, the Indus river system, the Ganges-Brahmaputra-Meghna (GBM) basin, and the shared aquifers of the Indo-Gangetic Plain — is the physical foundation of civilisation for approximately 1.8 billion people. The river systems that drain the Hindu Kush-Himalayan (HKH) water tower provide irrigation water for approximately 200 million hectares of South Asian farmland, drinking water for hundreds of millions of urban residents, hydropower for three nations (Nepal, Bhutan, India), and the ecological basis for the subcontinent's extraordinary biodiversity. This water architecture is under simultaneous stress from three converging forces: accelerating Himalayan glacier retreat driven by climate change, population and economic growth increasing per-capita freshwater demand, and geopolitical competition between India, Pakistan, China, and smaller riparian states that makes cooperative water management politically difficult even when hydrologically necessary.

The International Centre for Integrated Mountain Development (ICIMOD) 2023 assessment found that Hindu Kush Himalayan glaciers are losing mass at an accelerating rate — a 65% faster rate in the 2010s than in the previous decade — and that even under optimistic (1.5°C) warming scenarios, the HKH will lose one-third of its glacial ice by 2100. Under high-emission scenarios (4°C), 70-80% of glacial volume is lost. The practical implications are stark: the "Asian Water Tower" that feeds 12 of the world's major river systems — affecting 1.9 billion people — is melting. The initial effect is increased river flows (as glaciers melt faster); the medium-term effect (2040-2070) is reduced base flows as glacier volume declines; the long-term effect is seasonal flow disruption, more intense floods and droughts, and fundamental changes to the agricultural calendar of South Asian civilisation.

Part I — The Indus Water Treaty: 65 Years Old and Under Stress

The Treaty Architecture

The Indus Waters Treaty (IWT) — signed on September 19, 1960, between India and Pakistan, brokered by the World Bank — is one of the world's most consequential international water agreements. The treaty divided the Indus basin's six rivers between the two countries: India received exclusive use of the three "Eastern Rivers" (Sutlej, Beas, Ravi); Pakistan received exclusive use of the three "Western Rivers" (Indus, Jhelum, Chenab) — which carry approximately 80% of Indus basin total flow. India is permitted limited consumptive use of the Western Rivers for irrigation and run-of-river (non-storage) hydropower.

The treaty has survived two full-scale wars, three limited conflicts, multiple crises, and 65 years of often tense bilateral relations — making it, in theory, one of international water law's success stories. But the

treaty is showing structural strain:

Indian hydropower projects on Western Rivers: India's construction of run-of-river hydropower projects on the Jhelum (Kishanganga/Neelum) and Chenab (Ratle) rivers — permitted by the treaty but contested by Pakistan on design grounds — has been the subject of persistent arbitration. Pakistan contends these projects violate treaty storage provisions; India contends they comply with run-of-river specifications. The Permanent Court of Arbitration (PCA) and the International Court of Justice have both been invoked, creating parallel legal proceedings.

Indian IWT notification suspension: India suspended notifications to Pakistan under the IWT permanent Indus Commission mechanism in January 2023, citing Pakistan-based cross-border terrorism as justification for reconsidering treaty obligations. This was the first time India had formally threatened IWT review, and it signals a willingness to weaponise water as a geopolitical instrument that, if acted upon, would constitute a dangerous precedent for water law globally.

Climate change and treaty design: The IWT's fixed allocation (percentage of river flow) was designed for a stable climatic baseline. As glacial melt changes seasonal flow distributions, the treaty's technical assumptions about flow timing and volume may no longer hold, requiring fundamental renegotiation — a politically near-impossible task given current bilateral relations.

Part II — China's Upstream Power: The Brahmaputra Dam Scenario

China's Himalayan Water Leverage

China — which controls the Tibetan Plateau headwaters of the Brahmaputra (called the Yarlung Tsangpo in Tibet), the Mekong, the Salween, and the Indus — is upstream of virtually every major South and Southeast Asian river system. This geographic position gives China unparalleled potential leverage over downstream nations' water security: what China builds, diverts, or operates on its territory determines what flows to India, Bangladesh, Nepal, and Southeast Asia.

China's dam-building on the Brahmaputra has been the most geopolitically sensitive water issue in South Asia. China announced in 2020 that a dam at the Motuo/Great Bend of the Brahmaputra — where the river drops approximately 2,000 metres in one of the world's most powerful hydropower sites — would be included in China's 14th Five-Year Plan. The Motuo dam, when completed, will have a planned capacity of approximately 60 GW — three times the capacity of the Three Gorges Dam and the largest hydropower project in human history. From India's perspective, the Motuo dam (near the contested Arunachal Pradesh border) represents both a flood control risk (if water is released suddenly) and a water security risk (if China operates the dam to affect seasonal flows to downstream Assam and Bangladesh).

India has no formal bilateral water treaty with China on the Brahmaputra — only a 2013 Memorandum of Understanding to share hydrological data during flood season, which China has periodically withheld during political tensions. The absence of a treaty — combined with the scale of upstream dam development — means India and Bangladesh face an existential water security risk from an upstream power with which they have territorial disputes and no binding water agreement.

Part III — The Ganges: Sacred River, Stressed Ecosystem

India's Own Water Crisis

The Ganges — sacred to 1 billion Hindus, the river system supporting approximately 40% of India's population in its basin — is facing a crisis that is simultaneously ecological, social, and civilisational. The World Bank estimates that 40% of India's total water demand will exceed supply by 2030 under business-as-usual scenarios. Groundwater depletion in the Indo-Gangetic Plain — where farmers pump

irrigation water from aquifers that are recharged by the Ganges system — has been accelerating at 10-17 mm per year since 2002, as detected by NASA's GRACE satellite gravity measurements. The combination of over-extraction for irrigation, industrial and municipal pollution (the Ganges receives approximately 2.7 billion litres of domestic wastewater daily, primarily untreated), and glacial melt-driven flow changes creates a water security threat of exceptional scale.

Prime Minister Modi's Namami Gange programme — launched 2014 with Rs 20,000 crore (\$2.4 billion) commitment — has invested in sewage treatment plants, river surface cleaning, and public awareness. Progress is real but insufficient: approximately 55% of Ganga tributary towns now have sewage treatment coverage (versus 30% in 2014), but industrial effluent (tanneries, textile mills, pharmaceutical factories) remains inadequately controlled, and new agricultural runoff (fertilisers and pesticides) continues to create oxygen-depleted zones.

Part IV — Glacial Lake Outburst Floods: The Sudden Threat

The GLOF Danger

As Himalayan glaciers retreat, they leave behind glacial lakes — pools of meltwater contained by ice or moraine dams that can catastrophically fail, releasing walls of water, ice, and debris at speeds of up to 35 km/hour. Glacial Lake Outburst Floods (GLOFs) have increased in frequency in the Hindu Kush Himalaya region: the ICIMOD database records a significant increase in GLOF events since 2000. The most dramatic recent example: the Chamoli disaster of February 7, 2021 (Uttarakhand, India), where a glacier collapse and ice-rock avalanche triggered a flash flood that destroyed two hydropower projects and killed 200+ people in minutes.

For Bhutan — where glacial lakes have been catalogued and monitored since the 1990s — the GLOF risk is existential: a major GLOF from Thorthormi Lake (the largest glacial lake in Bhutan at 1.5 km² with an ice dam of uncertain stability) could cause catastrophic downstream flooding affecting entire river valleys. The Bhutanese government has implemented managed drainage of Thorthormi Lake (lowering the water level by 4+ metres through a controlled channel) — a globally-noted example of proactive GLOF risk management that required years of high-altitude engineering. For Nepal, Pakistan's Gilgit-Baltistan region (where 3,000+ glacial lakes exist), and India's Himalayan states (Uttarakhand, Himachal Pradesh, Sikkim), GLOF risk represents an unpredictable, potentially catastrophic threat to downstream populations and infrastructure.

Conclusion: Water as the Region's Existential Variable

South Asia's water crisis will be the most consequential geopolitical variable in the region by 2050 — surpassing terrorism, nuclear risk, and economic competition in its potential to destabilise states. The intersection of accelerating glacier retreat, population growth (South Asia will have 2.1 billion people by 2050), expanding agricultural water demand, and the geopolitical competition over upstream dam construction creates a scenario in which water scarcity becomes an existential national security issue for Pakistan (which depends on the Indus), Bangladesh (which depends on the Brahmaputra and Ganges), and India's northern states.

The diplomatic architecture — the Indus Waters Treaty (stressed), the bilateral data-sharing MoU with China (inadequate), and the absence of any multilateral South Asian water cooperation mechanism — is insufficient for the water governance challenge of the coming decades. Building multilateral water institutions in a region where nuclear-armed states have active territorial disputes and no tradition of supranational governance is extraordinarily difficult — but without it, the water conflicts of the 2030s and 2040s will be managed through unilateral national action, upstream dam construction, and periodic crisis

rather than cooperative management.

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All data from ICIMOD, World Bank, NASA GRACE, IARI, and cited sources.